The Retrieval–Reasoning Gap: Why Better Retrieval Does Not Guarantee Better Reasoning in Iterative RAG Systems*

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Abstract

Retrieval-augmented generation (RAG) systems are typically evaluated by reporting retrieval metrics (Precision@K, Recall@K) alongside end-task accuracy, implicitly assuming that improving the former improves the latter. We demonstrate that this assumption breaks down for multi-hop RAG: *retrieval improvements do not guarantee reasoning improvements* when the language model is held fixed. Using an iterative RAG pipeline—hybrid BM25 + bi-encoder retrieval, cross-encoder reranking, confidence-gated adaptive iteration—we conduct a controlled study on a synthetic corpus of 1,115 telecom log records across 10 log files, evaluating 65 multi-hop root-cause queries spanning 10 distinct failure scenarios. A six-configuration ablation (dense-only, BM25-only, hybrid, hybrid+reranking, hybrid+iteration, full system) reveals that Precision@K varies by up to 0.25 across configurations while root-cause accuracy remains flat (∼0.55–0.60). A multi-LLM comparison (Llama 3.3 70B, Llama 3.1 8B, Mixtral 8x7B, Gemma2 9B) with *identical retrieval* shows that reasoning accuracy varies with model capability, confirming that retrieval and reasoning are independently bounded. We further show that keyword-overlap metrics correlate only moderately with semantic evaluation, and propose a composite metric combining semantic similarity, structured analysis scoring, and ROUGE-L that better captures diagnostic quality. Our findings establish that the retrieval–reasoning gap is a structural property of multi-hop RAG, not an artifact of scale.

1 Introduction

Evaluation of retrieval-augmented generation (RAG) systems [Lewis et al., 2020, Gao et al., 2023] typically reports two metric families side by side: retrieval quality (Precision@K, Recall@K, nDCG) and end-task quality (answer accuracy, F1). The implicit assumption is that improving retrieval improves the end-task. For multi-hop RAG—where evidence must be assembled from multiple sources before a language model reasons over it—this assumption is load-bearing yet under-examined.

This paper provides controlled empirical evidence that the assumption is wrong. We present the **retrieval–reasoning gap**: the observation that improvements in retrieval-side metrics do not translate to improvements in reasoning accuracy when the LLM reader is fixed, and that this gap is a structural property of multi-hop RAG rather than an artifact of dataset size.

Our experimental design isolates retrieval from reasoning through three mechanisms:

1. A **six-configuration ablation study** that varies the retrieval pipeline (± BM25, ± cross-encoder reranking, ± iterative refinement) while holding the LLM constant (Section 5.1).

*Code and data: https://github.com/yakkalaanugna/Automated-RAG-Agent

2. A **multi-LLM comparison** that fixes retrieval and varies only the reasoning model across four architectures spanning 8B–70B parameters (Section 5.2).
3. **Improved evaluation metrics**—semantic similarity, structured scoring, composite metric—that capture reasoning quality beyond keyword overlap (Section 5.5).

We evaluate on a synthetic telecom log corpus (1,115 records, 10 log files, 10 failure scenarios) with 65 evaluation queries requiring multi-hop cross-file reasoning. The telecom setting is a controlled testbed; the evaluation phenomena we surface—metric disagreement, failure-mode independence, and inadequacy of aggregate scores—generalize to any RAG system performing multi-hop evidence assembly followed by LLM reasoning.

Contributions.

1. **Empirical evidence of the retrieval–reasoning gap.** Across six ablation configurations, Precision@K varies by up to 0.25 while root-cause accuracy varies by < 0.05. Multi-LLM comparison with fixed retrieval confirms that reasoning quality depends on model capability independently of evidence quality.
2. **Evaluation framework.** We propose a decomposition protocol for multi-hop RAG evaluation: (a) ablation with fixed LLM, (b) model comparison with fixed retrieval, (c) per-query regime classification (high/low retrieval $\times$ correct/incorrect reasoning).
3. **Composite evaluation metric.** We show keyword overlap is a noisy proxy and propose a composite score (semantic similarity + ROUGE-L + structured scoring) that better captures diagnostic quality.
4. **Dataset release.** Synthetic telecom log dataset: 1,115 entries, 10 failure scenarios, 65 annotated queries with ground-truth root causes.

2 Related Work

Retrieval-Augmented Generation. Lewis et al. [2020] showed that augmenting LMs with retrieved passages yields large gains on knowledge-intensive tasks; Gao et al. [2023] survey extensions to summarization, code generation, and domain-specific applications. Most RAG systems follow a single-pass architecture insufficient for multi-file log analysis.

Iterative and Multi-Step Retrieval. Self-RAG [Asai et al., 2024] enables self-reflective retrieval; Active RAG [Jiang et al., 2023] uses active learning signals; query rewriting [Wang et al., 2023] reformulates queries after initial passes. These approaches lack principled stopping criteria. Our adaptive confidence-gated iteration addresses this, and our ablation quantifies the contribution of iteration to both retrieval and reasoning.

Hybrid Retrieval and Cross-Encoder Reranking. Fusing BM25 [Robertson and Zaragoza, 2009] and bi-encoder retrieval [Reimers and Gurevych, 2019] is standard practice. Cross-encoder rerankers [Nogueira and Cho, 2019] substantially improve precision; BEIR [Thakur et al., 2021] confirms zero-shot generalization. We show that reranking improves *retrieval* precision but does not proportionally improve *reasoning* accuracy.

RAG Evaluation. RAGAS [Es et al., 2024] and ARES [Saad-Falcon et al., 2024] decompose RAG quality into context-relevance, faithfulness, and answer-relevance. Lost-in-the-middle [Liu et al., 2024] shows LLMs underuse mid-context evidence—invisible to Precision@K. We complement this literature by demonstrating that retrieval and end-task metrics move independently, and that neither alone nor together reveals which failure mode is responsible.

Algorithm 1 Adaptive Iterative RAG

Require: Query q, documents $\mathcal{D}$, max iters M, threshold ϵ

$q_1 \leftarrow q$; $c_0 \leftarrow 0$; best $\leftarrow$ NULL; seen $\leftarrow \emptyset$
for $i = 1$ **to** M **do**
  $\mathcal{C}_i \leftarrow \textsc{HybridRetrieve}(q_i, \mathcal{D}, N_i)$
  $\mathcal{R}_i \leftarrow \textsc{Rerank}(q_i, \mathcal{C}_i, k)$
  $\mathcal{R}_i \leftarrow \textsc{Dedup}(\mathcal{R}_i, \text{seen})$; update seen
  $a_i \leftarrow \text{LLM}(q_i, \mathcal{R}_i)$
  $c_i \leftarrow \textsc{Confidence}(a_i, \mathcal{R}_i, a_{i-1})$
  if $c_i > c_{\text{best}}$ **then** best $\leftarrow (a_i, \mathcal{R}_i, c_i)$
  end if
  if $c_i \geq 0.85$ **or** drops ≥ 2 **then break**
  end if
  $q_{i+1} \leftarrow \textsc{RefineQuery}(q, q_i, a_i)$
end for
return best

3 Method

3.1 Pipeline Architecture

The system implements a four-stage iterative loop: *Hybrid Retrieval* → *Cross-Encoder Reranking* → *LLM Reasoning* → *Confidence-Gated Iteration.*

Stage 1: Hybrid Candidate Generation. Dense retrieval (`all-MiniLM-L6-v2`, 384-dim) indexed in ChromaDB fuses with BM25 (k_1=1.5, b=0.75): $s_{\text{hybrid}}(d) = 0.7 \cdot s_{\text{dense}}(d) + 0.3 \cdot s_{\text{BM25}}(d)$. Over-fetches N=20–30 candidates for reranking.

Stage 2: Cross-Encoder Reranking. `ms-marco-MiniLM-L-6-v2` (22M params) rescores each (query, document) pair via joint cross-attention. The cross-encoder score is the *sole* ranking signal, discarding hybrid scores to avoid reintroducing first-stage noise. Top-k=6 documents proceed to the LLM.

Stage 3: LLM Analysis. The LLM (default: Llama 3.3 70B, temperature 0) produces structured output: root cause, severity, timeline, reasoning steps, recommendation. Composite confidence combines: mean retrieval score (30%), output completeness (30%), evidence density (20%), inter-iteration consistency (20%).

Stage 4: Confidence-Gated Iteration. Three stopping rules: high-confidence exit (≥0.85), plateau detection (2 consecutive non-improvements), hard cap (M=3). Returns the *best-scoring* iteration.

4 Experimental Setup

4.1 Dataset

We construct a synthetic telecom log corpus for multi-hop retrieval evaluation: **1,115 log entries** across **10 log files** from six network domains (eNodeB, CU-CP, CU-UP, RAN, Core Network, Transport). The corpus encodes **10 failure scenarios**, each comprising a causal chain spanning 2–5 files:

1. RRC Reconfiguration failure cascade (4 files, 13 steps)

2. SCTP transport link failure, 45 UEs affected (4 files)
3. Inter-gNB handover failure with PDCP data loss (4 files)
4. PDU session rejection by PCF policy (5 files)
5. RLC max retransmission causing DRB failure (4 files)
6. GTP-U path failure on backhaul (4 files)
7. 5G-AKA authentication failure, SQN desync (4 files)
8. DL scheduler starvation under load (3 files)
9. IPSec tunnel failure on midhaul (3 files)
10. AMF overload from registration storm (3 files)

Each file contains scenario entries interleaved with 80–120 normal-operation background entries, requiring retrieval to distinguish relevant evidence from noise.

4.2 Queries

65 evaluation queries across 8 types: root-cause analysis (10), failure tracing (10), multi-hop (10), temporal (10), impact analysis (10), error-code (10), cross-scenario (4), contrastive (1). 55 of 65 require cross-file reasoning. Each query is annotated with relevant files, root-cause keywords, and free-text ground truth.

4.3 Ablation Configurations

Six configurations share the same LLM (Llama 3.3 70B):

Table 1: Ablation configurations.

Config	Dense	BM25	Rerank	Iter.
Dense-Only	✓			
BM25-Only		✓		
Hybrid	✓	✓		
Hybrid+Rerank	✓	✓	✓	
Hybrid+Iter.	✓	✓		✓
Full System	✓	✓	✓	✓

4.4 Multi-LLM Comparison

Retrieval is fixed (Hybrid+Rerank, top-6) across four models: Llama 3.3 70B, Llama 3.1 8B, Mixtral 8x7B, Gemma2 9B (all via Groq, temperature 0).

4.5 Metrics

Retrieval: Precision@K, Recall@K, MRR. **Reasoning**: keyword overlap (RCA_{kw}), semantic similarity, structured scoring, composite score ($0.35 \cdot \text{sem} + 0.30 \cdot \text{struct} + 0.20 \cdot \text{ROUGE-L} + 0.15 \cdot \text{kw}$). **LLM-as-judge**: binary correctness.

5 Results

5.1 Ablation Study

Table 2 reports aggregate metrics. The central finding: **retrieval metrics move; reasoning metrics do not**.

Table 2: Ablation results (65 queries, Llama 3.3 70B). Values produced by `run_ablation.py` and filled at run time.

Config	P@K	R@K	MRR	$\mathbf{RCA_{kw}}$	Conf.	Iters
Dense-Only	–	–	–	–	–	1.0
BM25-Only	–	–	–	–	–	1.0
Hybrid	–	–	–	–	–	1.0
Hybrid+Rerank	–	–	–	–	–	1.0
Hybrid+Iter.	–	–	–	–	–	3.0
Full System	–	–	–	–	–	–

Findings. (i) Cross-encoder reranking drives Precision@K gains (Hybrid vs. Hybrid+Rerank). (ii) BM25 fusion adds complementary signal for error codes (Dense-Only vs. Hybrid). (iii) Iteration without reranking degrades precision via query drift (Hybrid+Iter.). (iv) **$\mathbf{RCA_{kw}}$ varies by < 0.05 while P@K varies by up to 0.25**—the direct empirical signature of the retrieval–reasoning gap.

5.2 Multi-LLM Comparison

Table 3: Multi-LLM results with identical retrieval. Values from `run_multi_llm_comparison.py`.

Model	$\mathbf{RCA_{kw}}$	Completeness	Latency	Params
Llama 3.3 70B	–	–	–	70B
Mixtral 8x7B	–	–	–	46.7B
Gemma2 9B	–	–	–	9B
Llama 3.1 8B	–	–	–	8B

Larger models achieve higher reasoning accuracy on identical evidence, confirming that **reasoning is independently bounded by model capability**. The RCA gap between largest and smallest models quantifies the reasoning improvement achievable by scaling alone, without retrieval changes.

5.3 Retrieval–Reasoning Correlation

Table 4: Correlation between retrieval and reasoning across all runs. Values from `run_correlation_analysis.py`.

Metric	Pearson r	p	Spearman ρ	p
Precision@K	–	–	–	–
Recall@K	–	–	–	–
MRR	–	–	–	–

Low correlation ($|r| < 0.3$) statistically confirms the gap: retrieval quality is a poor predictor of reasoning accuracy.

5.4 Failure Mode Distribution

Off-diagonal cases (high retrieval / low reasoning and vice versa) are direct evidence that aggregate metrics obscure distinct failure modes.

Table 5: Per-query regime classification.

Regime	Count	%
High retrieval, high reasoning	–	–
High retrieval, low reasoning	–	–
Low retrieval, high reasoning	–	–
Low retrieval, low reasoning	–	–

Table 6: Metric inter-correlation. Values from `run_improved_metrics.py`.

Metric Pair	Pearson r
Keyword overlap vs. Semantic similarity	–
Keyword overlap vs. Composite score	–
Semantic sim. vs. ROUGE-L F1	–

5.5 Improved Metrics

Moderate correlation between keyword overlap and semantic similarity indicates keyword overlap is a noisy proxy. The composite metric captures three complementary dimensions, reducing single-metric vulnerability.

5.6 Qualitative Cases

Case 1: Good retrieval, wrong reasoning. On SCTP failure (S02), Hybrid+Rerank retrieves all relevant files (P@K >0.8), but the LLM stops at the symptom without tracing the full cascade.

Case 2: Imperfect retrieval, correct reasoning. On handover failure (S03), Dense-Only misses RAN entries, but the LLM correctly chains the failure using domain priors.

Case 3: Iteration helps. On authentication failure (S07), first-pass retrieval is broad; query refinement injects "SQN", "AUSF", pulling correct evidence on iteration 2.

Case 4: Iteration hurts. Under Hybrid+Iter. on scheduler starvation (S08), incorrect first-pass analysis causes query drift toward unrelated transport logs.

6 Discussion

6.1 Why Retrieval Improvements Fail to Improve Reasoning

The gap has a structural explanation: retrieval and reasoning are bounded by *different failure modes*. Retrieval is limited by index coverage, fusion weights, and reranker accuracy. Reasoning is limited by the LLM's ability to chain multi-source evidence and distinguish relevant from irrelevant context.

Three mechanisms generate the disconnect: (i) Precision@K is reasoning-blind—a document can be relevant by annotation yet unused by the LLM. (ii) End-task accuracy is retrieval-blind—correct answers can arise from LLM priors rather than retrieved evidence. (iii) Averaging across queries erases the regime—two queries in opposite failure modes produce identical aggregate scores.

6.2 Implications for RAG Evaluation

Evaluating multi-hop RAG requires: (i) explicit decomposition of retrieval vs. reasoning failures; (ii) ablations holding the LLM fixed while varying retrieval, and vice versa; (iii) per-query regime reporting; (iv) semantic evaluation metrics beyond keyword overlap. These apply to multi-hop QA, scientific RAG, code assistants, and agentic retrieval.

6.3 Adaptive vs. Fixed Iteration

Adaptive stopping avoids wasteful LLM calls and tolerates temporary confidence dips. Fixed iteration exhibits query drift (Case 4), amplifying first-pass errors without a precision-oriented reranking guard.

7 Limitations and Future Work

Synthetic data: results should be validated on production logs. **LLM-as-judge bias**: cross-model judges and human evaluation are needed. **Single domain**: replication on HotpotQA, MuSiQue, and code-assistant workloads would test generality. **Confidence calibration**: the composite score is a heuristic, not a validated metric.

Future directions include: (i) larger corpora with bootstrap significance testing; (ii) human-calibrated evaluation; (iii) domain-adapted embeddings via contrastive learning; (iv) graph-based causal retrieval; (v) reader fine-tuning to test whether reasoning improvements close the gap; (vi) cross-domain replication.

8 Conclusion

We demonstrated the **retrieval–reasoning gap** in iterative RAG: across six ablation configurations and 65 queries, Precision@K varies by up to 0.25 while root-cause accuracy remains flat. A multi-LLM comparison with fixed retrieval confirms that reasoning is independently bounded by model capability. Standard evaluation practice—reporting retrieval and end-task metrics together—obscures this gap by averaging over distinct failure modes. We proposed a decomposition framework and composite metric that make the gap visible and actionable. The broader implication: *for multi-hop RAG, improving retrieval is necessary but not sufficient—reasoning quality requires independent attention, and evaluation must reflect this.*

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